

Indian Institute of Technology Dharwad
CS201L: Artificial Intelligence Laboratory
Lab 2: Data Visualization and Descriptive analytics
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Problem Statement 2

A real-world medical dataset containing hospital appointment records is provided in a CSV file (`noshowappointments.csv`). The dataset includes various attributes that provide insights into patient appointment behavior. These features are:

Table 1: Description of Attributes in the Patient Appointment Dataset

Feature	Description
PatientId	Unique identifier assigned to each patient.
AppointmentID	Unique identifier assigned to each appointment record.
Gender	Gender of the patient.
ScheduledDay	Date on which the appointment was scheduled.
AppointmentDay	Actual date of the appointment.
Age	Age of the patient in years.
Neighbourhood	Residential area where the patient lives.
Scholarship	Indicates whether the patient is enrolled in a social welfare program.
Hypertension	Indicates whether the patient has hypertension.
Diabetes	Indicates whether the patient has diabetes.
Alcoholism	Indicates whether the patient has a history of alcoholism.
SMS_received	Indicates whether the patient received a reminder SMS.
No-show	Indicates whether the patient failed to attend the appointment.

Write a Python program to perform the following tasks:

1. Display the five-number summary (Minimum, First Quartile Q_1 , Median Q_2 , Third Quartile Q_3 , and Maximum) of the **Age** attribute using the `describe()` function.
2. Compute the first quartile (Q_1), median (Q_2), third quartile (Q_3), and the interquartile range (IQR) for the **Age** attribute using the `quantile()` function.
3. Find the lower and upper bounds for the **Age** attribute using the Interquartile Range (IQR) rule, where

$$\text{Lower Bound} = Q_1 - 1.5 \times \text{IQR}, \quad \text{Upper Bound} = Q_3 + 1.5 \times \text{IQR}.$$

Compute these bounds using the `quantile()` function.

4. Determine the percentage of patients whose **Age** lies outside the interquartile range (IQR) bounds using the IQR rule.
5. Generate scatter plots to explore relationships between **Age**, **SMS_received**, **Scholarship**, and **AppointmentDay** using both Matplotlib and Seaborn libraries.
6. Plot histograms and Kernel Density Estimation (KDE) curves for all numerical features using Matplotlib and Seaborn, with superimposed mean, median, and $\pm 2.7\sigma$ control limit lines and annotated values to facilitate comprehensive distribution analysis.
7. Group the data according to the attribute **No-show** and generate bar plots to analyze the distribution of **Age**, **SMS_received**, and **Scholarship** using both Matplotlib and Seaborn libraries.
8. Create boxplots for all numerical attributes using both Matplotlib and Seaborn libraries.